Project Abstract
Placement Prediction with AI Resume Optimizer

Track	Edutech & Learning Innovation
Problem Statement	Create an ML-based tool that predicts placement probability based on student profile and suggests improvements in resume, skill-building, and project portfolio
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Abstract

Campus placement is one of the most high-stakes milestones in a student's academic journey — yet most students enter the process blind, with no data-driven understanding of their employability or a clear roadmap to improve it. This project presents **Placement Prediction with AI Resume Optimizer**, a dual-purpose career readiness platform designed to solve this challenge end-to-end using Machine Learning, NLP, and LLM-powered resume optimization.

Problem Statement

Engineering students today face three silent failures that cost them years of effort and opportunity.
Resume Rejected Before HR Ever Reads It. Over 75% of resumes are filtered out automatically by Applicant Tracking Systems due to poor keyword alignment, wrong formatting, and missing role-specific skills. The student never receives a rejection reason. They reapply with the same flawed resume, get rejected again, and never understand why. A single formatting mistake or missing keyword can end a candidacy silently before any human is involved.
Students Cannot Predict Where They Stand. Most students go through their first, second, and third year of college with no understanding of their placement readiness. They do not know which company tier they qualify for, what technical skills are missing, which extracurricular activities strengthen a profile, or whether their current CGPA and project portfolio is enough. By the time placement season starts in the final year it is too late to close these gaps. Four years of effort can fail because of six months of blindness.

Proposed Solution

AIRO addresses each failure directly through two integrated systems:

Part 1 — Placement Prediction and Gap Analysis
An **XGBoost** classifier trained on 10,000+ student records predicts each student's placement probability from 0 to 100% based on CGPA, skills, internships, projects, backlogs, and certifications. A **Random Forest** model then predicts the company tier the student is most likely to be placed in — FAANG, Mid-tier, or Mass Recruiter. **SHAP** (SHapley Additive Explanations) breaks down every prediction into individual feature contributions, showing the student exactly which skills are their strengths, which gaps are hurting their score, and what extracurricular or technical areas they need to develop — from Year 1, not just in the final semester.

Part 2 — Resume Optimization and Tailoring
The student uploads their existing resume and a target job description. **Sentence Transformers** compute a semantic similarity score between each resume section and the JD requirements, identifying which parts are weak or misaligned. A **Groq LLM** then rewrites those sections intelligently to match the role — not by stuffing keywords but by capturing role-specific context and tone. The student sees their ATS score before and after tailoring so the improvement is measurable. The final optimized resume is exported as a professional PDF via **XeTeX**, turning a 45-minute manual process into under 60 seconds.

Technology Stack

Technology	Role	Technology	Role
Next.js + React	Frontend	**FastAPI + Python**	Backend
XGBoost	Placement probability prediction	**Random Forest**	Company tier classification
SHAP	Prediction explainability	**Sentence-Transformers**	Resume–JD similarity scoring
Groq LLM API	Resume rewriting	**MongoDB Atlas**	Database
XeTeX / LaTeX	PDF resume generation	**JWT + OAuth2**	Authentication
FAISS	Semantic search	**Docker + Render.com**	Deployment

Expected Output & Impact

The platform delivers a Placement Probability Score with SHAP-powered feature breakdown, a Company Tier prediction (FAANG / Mid-tier / Mass Recruiter / Not Placed), a personalised skill improvement roadmap, and a professionally formatted ATS-optimized LaTeX PDF resume tailored to the student's target Job Description. Target beneficiaries include engineering students preparing for campus placements, career development cells, and placement coordinators seeking data-driven support tools.

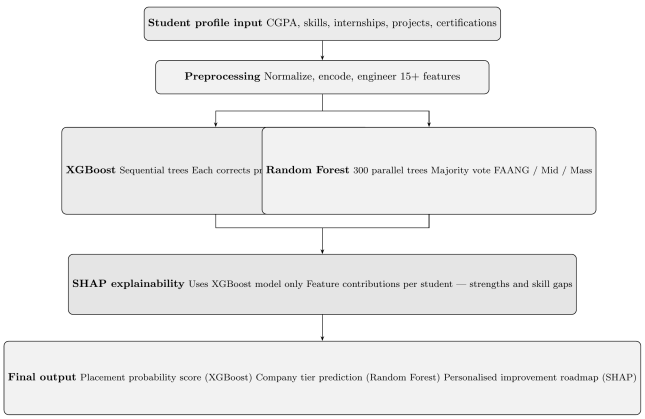

Figure 1: ML Model Pipeline

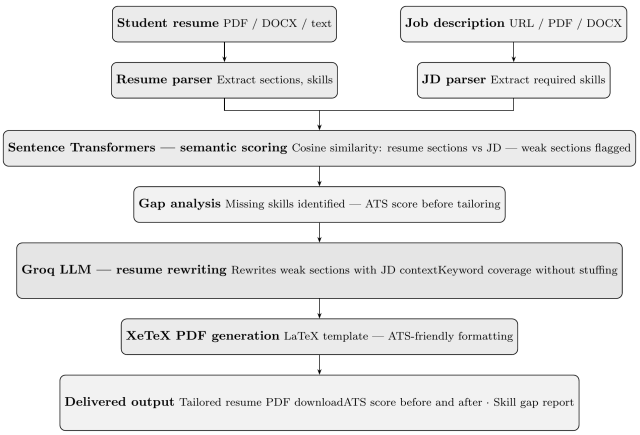

Figure 2: Resume Tailoring Pipeline